Problem 6.5

You invest some money in an account earning 6% annual compound interest. How long will it take to quadruple your account balance? (Express your answer in years to two decimal places.)

We use the compound interest formula:

$$A = P(1 + r)^t.$$

To quadruple the balance, we set $A = 4P$, so

$$4P = P(1.06)^t.$$

Dividing both sides by P,

$$4 = (1.06)^t.$$

Taking natural logarithms on both sides,

$$\ln(4) = t \ln(1.06).$$

Solving for t,

$$t = \frac{\ln(4)}{\ln(1.06)}.$$

Therefore,

$$t \approx 23.79 \text{ years}.$$

So, it will take approximately

$$\boxed{23.79 \text{ years}}.$$